

Gaurav Singh

STRUCTURAL & COMPUTATIONAL ENGINEER

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[Portfolio](#) · [GitHub](#) · [LinkedIn](#)

Structural engineer and scientific software developer (M.Tech, IIT Gandhinagar) specialising in turning design codes into validated, production-grade engineering software. 2+ years building commercial structural analysis and design tools used by engineers across India and South Korea, with deep expertise in nonlinear FEM, composite steel-concrete design, and automated verification. Fluent across the full computational-mechanics stack — from CDP-based Abaqus simulations and MATLAB optimisation frameworks to VB.NET/C# module development and Python verification pipelines.

EDUCATION

M.Tech, Structural Engineering

Indian Institute of Technology (IIT) Gandhinagar

CPI 7.5 · 2023 – 2025

B.Tech, Civil Engineering

Indian Institute of Technology (IIT) Palakkad

CGPA 7.0 · 2017 – 2021

PROFESSIONAL EXPERIENCE

Technical Specialist

Jun 2026 – Present

Acqisy Solutions Pvt. Ltd. — Siemens AEC (S-FRAME, S-CONCRETE, S-FOUNDATION)

- Develop composite-section design and analysis modules for commercial structural software, with automated test suites that validate engineering results on every release.
- Built a **Python verification framework** that auto-generates S-FOUNDATION input models, runs independent engineering calculations, compares outputs, and produces LaTeX verification reports.
- Conducted comparative studies of SRC and CFT composite-member provisions across **IS 11384, AISC, and KDS**; verified column designs for axial–flexural interaction, shear, slenderness, and buckling via strain-compatibility and interaction-curve methods.
- Implemented structural design algorithms in VB.NET (engineering calculations, I/O handling, report generation) and authored benchmark examples supporting QA and technical documentation.

R&D Engineer

Sep 2025 – Jun 2026

Modulus Housing Pvt. Ltd.

- Led R&D on cost optimisation of RC elements (beams, columns, footings, wall panels) using Abaqus and custom code, achieving major material and cost savings over conventional prismatic members.
- Built a MATLAB nonlinear beam-optimisation framework (IS 456) using segment-wise optimisation driven by bending-moment and shear distributions — cut element cost **40–50%** while meeting strength and serviceability limits.
- Ran nonlinear FEA with the **Concrete Damage Plasticity** model (material, geometric, and contact nonlinearities); simulated modular RC bathroom pods under crane-lift, transport, vibration, modal, and impact loads to certify handling safety.
- Validated TRC panels, EPS wall systems, and UHPC/FRC members using fiber-model material generators and research benchmarks; designed manufacturable parametric CAD models and casting moulds in SolidWorks for fabrication and testing.
- Drove global structural analysis by converting AutoCAD drawings into STAAD.Pro models and extracting force envelopes to feed optimisation workflows.

CAX Software Engineer

Jul 2024 – Sep 2025

Acqisy Solutions Pvt. Ltd. — Altair AEC (S-FRAME, S-CONCRETE, S-FOUNDATION)

- Developed and validated structural design calculation frameworks (beams, columns, slabs, walls) in Maple Flow; cross-checked against ETABS, STAAD.Pro, textbooks, and Excel tools.
- Implemented **IS 456, IS 13920, and KDS 14 20** reinforced-concrete modules in S-CONCRETE using VB.NET and C#, enabling software adoption by engineers across India and South Korea.
- Led to S-FOUNDATION development alongside its finite element solver — validated rigid and flexible finite-element formulations for isolated, combined, raft, wall, and pile foundations with soil–structure interaction against hand calculations and benchmarks, developed finite element solver main engine in C# for analysis, brainstormed ideas for future developments additions to the software.

- Improved S-FRAME interoperability with STAAD.Pro, ETABS, and MIDAS by reverse-engineering proprietary file formats and building import/export pipelines in C#/WPF; delivered design reports, UI/UX, and end-user documentation.
- Collaborated with Altair Engineering (Canada) and Korean engineering partners to align features with regional codes and practice.

Project Associate (part-time, alongside M.Tech)

Jan 2023 – Jul 2024

Indian Institute of Technology (IIT) Gandhinagar

- Built fire-dynamics and pedestrian-evacuation simulations (FDS+Evac) of Bengaluru metro stations from detailed CAD layouts to assess emergency egress under fire scenarios — industry consultancy project.
- Automated **150+ parametric simulation scenarios** via custom MATLAB scripting; applied Design of Experiments and Explainable ML to quantify the influence of **15+ fire and occupant parameters** on evacuation time, bottlenecks, smoke spread, and congestion.
- Calibrated occupant-movement models to realistic Indian crowd behaviour; published as M.Tech thesis and presented at ICCMS 2023.

KEY PROJECTS & TECHNICAL WORK

- **Topology Optimisation of 2D Plates with Large Deformations** (*Academic*) — Built a nonlinear FEM topology-optimisation code (SIMP method) in MATLAB for Mindlin plates — 5 DOF/node, bilinear shape functions, geometric nonlinearity via Newton–Raphson — to study large-deformation effects on optimal material layout.
- **Design & Analysis of a G+5 Building** (*Academic*) — Performed structural analysis in SAP2000 and MATLAB with wind (IS 875) and seismic (IS 1893) loads; designed beams, columns, slabs, shear walls, and footings per IS 456 and IS 13920.
- **Experimental & Numerical Study of 3D Truss Structures** (*Academic*) — Fabricated and load-tested a 3D truss; built a MATLAB stiffness-matrix solver with substructuring and static condensation, validated against OpenSees.
- **Dynamic Analysis of MDOF Structures** (*Academic*) — Designed a 2-DOF structure with Arduino-controlled excitation; measured and filtered the response and validated a Newmark- β numerical model against OpenSees.
- **Engineering & Scientific Software** (*Personal*) — Built various tools such as NL-FEA tool, CFD flow simulations and Multiphysics solvers using (PyQt6 + VTK), various other learning apps such as a scientific calculator and virtual piano (Tkinter); regularly develop and open-source scientific and web tools on GitHub.

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, C#, VB.NET, C, HTML/JavaScript

Simulation: Abaqus, ANSYS, SAP2000, ETABS, STAAD.Pro, S-FRAME / S-CONCRETE / S-FOUNDATION, FDS+Evac, Mapleflow, Pyrosim, Pathfinder, open sources softwares like Openfoam, OpenSees and Fenicsx

CAD & Modelling: SolidWorks, AutoCAD, Revit

Courses and Methods: Nonlinear FEM, CFD, turbulence and combustion modelling, heat transfer, topology optimisation, composite steel-concrete design, fire engineering, Design of Experiments, machine learning, numerical methods, engineering optimization techniques

Codes & Standards: IS 456, IS 11384, IS 13920, IS 875, IS 1893, IS800, ACI 318, Eurocode 2, KDS 14 20, KDS 14 30, AISC 360 and ACI318

Other Tools: Maple Flow, SageMath, Minitab, Primavera, LaTeX, Git, Github

PUBLICATIONS & PRESENTATIONS

- “*Numerical Investigations on Evacuation Performance of Passengers in Underground Metro Stations*” — presented at the 9th International Congress on Computational Mechanics and Simulation (ICCMS), IIT Gandhinagar, Dec 2023; findings also presented to metro-industry professionals at an IIT Gandhinagar workshop (Jan 2024).

LEADERSHIP & VOLUNTEERING

- **eVidyaloka Teaching Volunteer** — taught mathematics to underprivileged rural students remotely and built educational content during COVID-19.
- **IITGN Neev Volunteer** — ran MS Office and computer-hardware training sessions for residents of Basan village.
- **Chegg Subject-Matter Expert** — solved and explained civil engineering problems for students worldwide.
- **Unnat Bharat Abhiyan** — surveyed rural residential complexes in Kerala to gather data for development projects.

INTERESTS

Football, sketching, and guitar. Writes a personal engineering blog on mathematics, physics, and the history of science.